Stochastic optimal control approach to generative modelling and Schrödinger potential estimation

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Steklov Mathematical Institute, June 8, 2026

Diffusion-based generative models

$$X_0 \sim \mathrm{p}_0 \qquad \mathrm{d}X_t = f(X_t, t)\,\mathrm{d}t + g(X_t, t)\mathrm{d}W_t \qquad X_T \sim \mathrm{p}_T$$



Figure: example of generating an image of cat from Gaussian noise

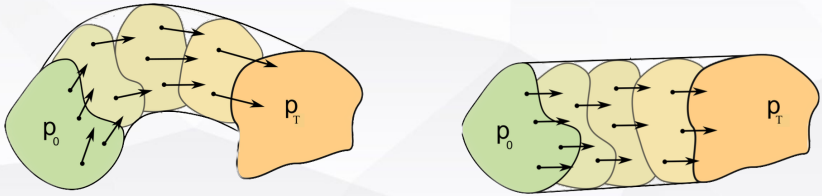
The drift $f(x, t)$ and volatility $g(x, t)$ are learnt from i.i.d. samples $Y_1, \ldots, Y_n \sim \mathrm{p}_T$

Diffusion-based generative models

$$X_0 \sim p_0 \qquad dX_t = f(X_t, t)\,dt + g(X_t, t)\,dW_t \qquad X_T \sim p_T$$

Observation 1: there exist multiple ways to transfer p_0 to p_T

Observation 2: simpler trajectories are easier to simulate



Little detour: Schrödinger bridge problem

Consider a diffusion process

$$\mathrm{d}X_t = f(t, X_t)\mathrm{d}t + g(t, X_t)\mathrm{d}W_t, \tag{1}$$

over the Euclidean space $\mathbb{R}^n$.

[Schrödinger, 1932]: a large number N of trajectories over the time interval $[0, 1]$ are independently sampled from (1) with $X_0 \sim \mathrm{p}_0$. LLN dictates that the terminal distribution at time $t = 1$ must be (approximately)

$$\int_{\mathbb{R}^n} q(0, x, 1, \cdot)\mathrm{p}_0(x)\, dx, \tag{2}$$

where $q(t, x, s, y)$, $t < s$ denotes the transition kernel. Suppose the observed marginal distribution at time $t = 1$, denoted by p_1, is inconsistent with (2) and the prior kernel $q(0, x, 1, y)$, that is,

$$\mathrm{p}_1(\cdot) \neq \int_{\mathbb{R}^n} q(0, x, 1, \cdot)\mathrm{p}_0(x)\, dx.$$

Little detour: Schrödinger bridge problem

Let $\Omega = C([0,1], \mathbb{R}^n)$ be the space of continuous functions on $[0,1]$ with values in $\mathbb{R}^n$, and $\mathcal{P}(\Omega)$ denote the space of probability laws over Ω.

Consider N independent trajectories $X_t^1, X_t^2, \ldots, X_t^N \in \Omega$ of a diffusion having law $Q \in \mathcal{P}(\Omega)$, and let L_N denote their empirical distribution. Then, asymptotically as $N \to \infty$, Sanov's theorem gives:

$$\Pr(L_N \in A) \approx \exp\left(-N \inf_{P \in A} \mathrm{KL}(P\,,\,Q)\right), \quad \forall A \subset \mathcal{P}(\Omega), \tag{2}$$

where $\mathrm{KL}(P\,,\,Q) := \int \log\left(\frac{dP}{dQ}\right) dP$.

Problem 1. Let $Q \in \mathcal{P}(\Omega)$ be the probability measure on Ω induced by the prior process (1) with initial distribution p_0. Determine

$$P^\star := \arg \min_{P \in \mathcal{P}(\Omega)} \left\{ \mathrm{KL}(P,\, Q) \mid P_0 = p_0,\, P_1 = p_1 \right\}, \tag{3}$$

where P_t denotes the marginal distribution of P at time t (i.e., the push forward $X_t \# P = P_t$).

By disintegration of measure,

$$Q(\cdot) = \int_{\mathbb{R}^n \times \mathbb{R}^n} Q^{xy}(\cdot) Q_{01}(dx\, dy), \quad \text{and} \quad P(\cdot) = \int_{\mathbb{R}^n \times \mathbb{R}^n} P^{xy}(\cdot) P_{01}(dx\, dy),$$

where Q_{01} (P_{01}) denotes the joint marginal distribution of Q (P) of X_t for $t \in \{0, 1\}$ (i.e., $Q_{01} = X_{01}\#Q$, and similarly, for P), and Q^{xy} (P^{xy}) denotes the measure induced by P conditioned on $(X_0 = x, X_1 = y)$. It follows that

$$\mathsf{KL}(P\,,\, Q) = \mathsf{KL}(P_{01}\,,\, Q_{01}) + \int \mathsf{KL}(P^{xy}\,,\, Q^{xy}) P_{01}(dx\, dy). \tag{4}$$

Clearly, when $P^{xy} = Q^{xy}$ for any $x, y \in \mathbb{R}^n$, the second term on the right assumes the minimal value 0. An immediate consequence is the following *static* formulation of the SBP.

Problem 2. *Determine*

$$\pi^\star := \arg \min_{\pi \in \mathcal{P}(\mathbb{R}^n \times \mathbb{R}^n)} \left\{ \mathrm{KL}(\pi\,,\, Q_{01}) \mid \pi_0 = \rho_0,\, \pi_1 = \rho_1 \right\}. \tag{5}$$

Theorem: [Leonard, 2014]

Suppose $P^\star$ is a solution to the dynamic SBP (**Problem 1**), then $P^\star_{01}$ solves the static SBP (**Problem 2**). On the other hand, if $\pi^\star$ is a solution to (**Problem 2**), then setting

$$P^\star = \int_{\mathbb{R}^n \times \mathbb{R}^n} Q^{xy}(\cdot)\, \pi^\star(dx\, dy)$$

solves (**Problem 1**), while $P^\star_{01} = \pi^\star$.

Stochastic optimal control approach to generative modelling

Base process:

$$X_0^0 \sim \mathrm{p}_0, \quad \mathrm{d}X_t^0 = b(m - X_t^0)\,\mathrm{d}t + \Sigma^{1/2}\,\mathrm{d}W_t, \quad 0 < t < T$$

Controlled process:

$$X_t^u \sim \mathrm{p}_0, \quad \mathrm{d}X_t^u = b(m - X_t^u)\,\mathrm{d}t + u(X_t^u, t)\,\mathrm{d}t + \Sigma^{1/2}\,\mathrm{d}W_t, \quad 0 < t < T$$

Stochastic optimal control (SOC) problem with a final constraint:

$$\begin{cases} \int\limits_0^T \left\|\Sigma^{-1/2} u(X_t^u, t)\right\|^2 \mathrm{d}t \quad \longrightarrow \quad \min\limits_u \\ X_t^u \sim \mathrm{p}_0, \quad X_T^u \sim \mathrm{p}_T, \\ \mathrm{d}X_t^u = b(m - X_t^u)\,\mathrm{d}t + u(X_t^u, t)\,\mathrm{d}t + \Sigma^{1/2}\,\mathrm{d}W_t, \quad 0 < t < T \end{cases}$$

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Solution [Dai Pra, 1991]: $u^*(x,t) = \nabla \log \mathcal{T}_{T-t}[e^{\varphi^*}]$, where

- $\mathcal{T}_t$ is the Ornstein-Uhlenbeck operator: $\mathcal{T}_t[g](x) = \mathbb{E}\left[g(X_t^0) \,\middle|\, X_0^0 = x \right]$
- φ^* is a (right) Schrödinger potential

Theorem: [Beurling, 1960], in this form: [Dai Pra, 1991]

Let $q_T(y \mid x)$ stand for the conditional density of X_T^0 given $X_0^0 = x$. Then there exist unique (up to additive constants) ψ^* and φ^* such that the joint density

$$\pi^*(x, y) = e^{\psi^*(x)} q_T(y \mid x) e^{\varphi^*(y)}$$

has marginals $p_0(x)$ and $p_T(y)$

Remark 1. e^{ψ^*} and e^{φ^*} are referred to as (left and right) *Schrödinger potentials*

Remark 2. It is enough to estimate the (right) Schrödinger log-potential φ^* to learn the optimal control $u^*(x, t)$

Schrödinger equations:

$$\begin{cases} p_0(x) = \int\limits_{\mathbb{R}^d} \pi(x, y)\, dy = e^{\psi^*(x)} \int\limits_{\mathbb{R}^d} q_T(y \mid x) e^{\varphi^*(y)}\, dy \\ p_T(y) = \int\limits_{\mathbb{R}^d} \pi(x, y)\, dx = e^{\varphi^*(y)} \int\limits_{\mathbb{R}^d} q_T(y \mid x) e^{\psi^*(x)}\, dx \end{cases}$$

Remark. The left potential can be eliminated:

$$p_0(x) = e^{\psi^*(x)} \mathcal{T}_T[e^{\varphi^*}](x) \qquad \Longrightarrow \qquad e^{\psi^*(x)} = \frac{p_0(x)}{\mathcal{T}_T[e^{\varphi^*}](x)}$$

Entropic optimal transport (EOT):

$$\begin{cases} \int\limits_{\mathbb{R}^d}\int\limits_{\mathbb{R}^d} \|x-y\|^2 \pi(x,y)\,\mathrm{d}x\mathrm{d}y + 2\varepsilon\,\mathsf{KL}(\pi, \mathrm{p}_0 \otimes \mathrm{p}_T) \quad \longrightarrow \quad \min\limits_{\pi} \\ \int\limits_{\mathbb{R}^d} \pi(x,y)\,\mathrm{d}y = \mathrm{p}_0(x), \quad \int\limits_{\mathbb{R}^d} \pi(x,y)\,\mathrm{d}x = \mathrm{p}_T(x) \end{cases}$$

Theorem: [Leonard, 2014, Theorem 2.4]

The optimal coupling π° has a form

$$\pi^\circ(x,y) = e^{\psi^\circ(x)} \cdot e^{-\|x-y\|^2/(2\varepsilon)} \cdot e^{\varphi^\circ(y)}$$

Remark. The log-potentials ψ^* and φ^* coincide with ψ° and φ°, respectively, when $b = 0$, $m = 0$, $\Sigma = \varepsilon I_d$ (i.e. the base process is a scaled Brownian motion):

$$\mathrm{d}X_t^0 = \sqrt{\varepsilon}\,\mathrm{d}W_t, \quad 0 < t < T$$

Stochastic optimal control (SOC) problem with a final constraint:

$$\begin{cases} \int\limits_0^T \left\| \Sigma^{-1/2} u(X_t^u, t) \right\|^2 \, \mathrm{d}t \quad \longrightarrow \quad \min\limits_u \\ X_t^u \sim \mathrm{p}_0, \quad X_T^u \sim \mathrm{p}_T, \\ \mathrm{d}X_t^u = b(m - X_t^u)\,\mathrm{d}t + u(X_t^u, t)\,\mathrm{d}t + \Sigma^{1/2}\,\mathrm{d}W_t, \quad 0 < t < T \end{cases}$$

Advantages of stochastic optimal control framework:

- Flexible choice of the base process $\{X_t^0 : 0 \leqslant t \leqslant T\}$
- The matrix Σ helps us to take data anisotropy into account
- If $b > 0$, then the base process $\{X_t^0 : 0 \leqslant t \leqslant T\}$ is geometrically ergodic (in contrast to the Wiener process in EOT)

Controlled process with Schrödinger log-potential φ:

$$X_t^{\varphi} \sim \mathrm{p}_0, \quad \mathrm{d}X_t^{\varphi} = \big(bm - bX_t^{\varphi} + \Sigma\,\nabla \log \mathcal{T}_{T-t}[e^{\varphi}](X_t^{\varphi})\big)\,\mathrm{d}t + \Sigma^{1/2}\,\mathrm{d}W_t, \quad 0 < t < T$$

Remark: the joint density of X_0^{φ} and X_T^{φ} has a form

$$\pi^{\varphi}(x, y) = \frac{\mathrm{p}_0(x)\,\mathrm{q}_T(y \mid x)\,e^{\varphi(y)}}{\mathcal{T}_T[e^{\varphi}](x)}, \quad \mathrm{q}_T(y \mid x) - \text{density of } \mathcal{N}(m_t(x), \Sigma_t),$$

where

$$m_t(x) = (1 - e^{-bt})m + e^{-bt}x, \qquad \Sigma_t = (1 - e^{-2bt})\Sigma/(2b)$$

The marginal density of X_T^{φ} is given by

$$\mathrm{p}_T^{\varphi}(y) = \int \pi^{\varphi}(x, y)\,\mathrm{d}x = \int \frac{\mathrm{p}_0(x)\,\mathrm{q}_T(y \mid x)\,e^{\varphi(y)}}{\mathcal{T}_T[e^{\varphi}](x)}\,\mathrm{d}x$$

How to learn Schrödinger log-potential?

Strategy A: minimize KL-divergence between endpoint marginals

$$\mathrm{KL}(\mathrm{p}_T, \mathrm{p}_T^{\varphi}) \quad \longrightarrow \quad \min_{\varphi \in \Phi}$$

Strategy B: minimize KL-divergence between couplings

$$\mathrm{KL}(\pi^*, \pi^{\varphi}) \quad \longrightarrow \quad \min_{\varphi \in \Phi}$$

Here Φ is a reference class of log-potentials (e.g., neural networks) and

$$\pi^*(x, y) = \frac{\mathrm{p}_0(x)\, \mathrm{q}_T(y \mid x)\, e^{\varphi^*(y)}}{\mathcal{T}_T[e^{\varphi^*}](x)}, \quad \mathrm{p}_T(y) = \int \frac{\mathrm{p}_0(x)\, \mathrm{q}_T(y \mid x)\, e^{\varphi^*(y)}}{\mathcal{T}_T[e^{\varphi^*}](x)}\, \mathrm{d}x$$

- We have i.i.d. samples $Y_1, \dots, Y_n \sim \mathrm{p}_T$
- The source density p_0 is available, the target one p_T is unknown

Strategy A:

$$\mathrm{KL}(\mathrm{p}_T, \mathrm{p}_T^{\varphi}) \quad \to \quad \min_{\varphi \in \Phi} \qquad \Longleftrightarrow \qquad -\mathbb{E}_{Y \sim \mathrm{p}_T} \log \mathrm{p}_T^{\varphi}(Y) \quad \to \quad \min_{\varphi \in \Phi}$$

Empirical risk minimization:

$$\widetilde{\varphi} \in \operatorname*{Argmin}_{\varphi \in \Phi} \left\{ -\frac{1}{n} \sum_{i=1}^{n} \log \mathrm{p}_T^{\varphi}(Y_i) \right\}, \quad \text{where} \quad \mathrm{p}_T^{\varphi}(y) = \int \frac{\mathrm{p}_0(x)\, \mathrm{q}_T(y \mid x)\, e^{\varphi(y)}}{\mathcal{T}_T[e^{\varphi}](x)}\, \mathrm{d}x$$

- We have i.i.d. samples $Y_1, \dots, Y_n \sim \mathrm{p}_T$
- The source density p_0 is available, the target one p_T is unknown

Observation:

$$\mathrm{KL}(\pi^*, \pi^\varphi) = \int\limits_{\mathbb{R}^d} \int\limits_{\mathbb{R}^d} \log \frac{\pi^*(x,y)}{\pi^\varphi(x,y)}\, \pi^*(x,y)\, \mathrm{d}x\mathrm{d}y$$

$$= \int\limits_{\mathbb{R}^d} \int\limits_{\mathbb{R}^d} \log \left(\frac{\mathrm{p}_0(x)\, \mathrm{q}_T(y \mid x)\, e^{\varphi^*(y)}}{\mathcal{T}_T[e^{\varphi^*}](x)} \cdot \frac{\mathcal{T}_T[e^{\varphi}](x)}{\mathrm{p}_0(x)\, \mathrm{q}_T(y \mid x)\, e^{\varphi(y)}} \right) \pi^*(x,y)\, \mathrm{d}x\mathrm{d}y$$

$$= \int\limits_{\mathbb{R}^d} \log \frac{\mathcal{T}_T[e^{\varphi}](x)}{\mathcal{T}_T[e^{\varphi^*}](x)} \underbrace{\left(\int\limits_{\mathbb{R}^d} \pi^*(x,y)\, \mathrm{d}y \right)}_{\mathrm{p}_0(x)} \mathrm{d}x - \int\limits_{\mathbb{R}^d} \big(\varphi(y) - \varphi^*(y)\big) \underbrace{\left(\int\limits_{\mathbb{R}^d} \pi^*(x,y)\, \mathrm{d}x \right)}_{\mathrm{p}_T(y)} \mathrm{d}y$$

- We have i.i.d. samples $Y_1, \dots, Y_n \sim \mathrm{p}_T$ and $Z_1, \dots, Z_n \sim \mathrm{p}_0$
- The source density p_0 is available, the target one p_T is unknown

Observation:

$$\mathrm{KL}(\pi^*, \pi^\varphi) = \mathcal{L}(\varphi) - \mathcal{L}(\varphi^*), \quad \text{where} \quad \mathcal{L}(\varphi) = \mathbb{E}_{Z\sim \mathrm{p}_0} \log \mathcal{T}_T[e^\varphi](Z) - \mathbb{E}_{Y\sim \mathrm{p}_T}\varphi(Y)$$

Strategy B:

$$\mathrm{KL}(\pi^*, \pi^\varphi) \quad \to \quad \min_{\varphi\in\Phi} \quad \Longleftrightarrow \quad \mathcal{L}(\varphi) \quad \to \quad \min_{\varphi\in\Phi}$$

Empirical risk minimization:

$$\widehat{\varphi} \in \operatorname*{Argmin}_{\varphi\in\Phi} \left\{ \frac{1}{n}\sum_{j=1}^{n} \log \mathcal{T}_T[e^\varphi](Z_j) - \frac{1}{n}\sum_{i=1}^{n} \varphi(Y_i) \right\}$$

$$\widetilde{\varphi} \in \operatorname*{Argmin}_{\varphi \in \Phi} \left\{ -\frac{1}{n} \sum_{i=1}^{n} \log \mathrm{p}_T^{\varphi}(Y_i) \right\} \quad \text{vs} \quad \widehat{\varphi} \in \operatorname*{Argmin}_{\varphi \in \Phi} \left\{ \frac{1}{n} \sum_{j=1}^{n} \log \mathcal{T}_T[e^{\varphi}](Z_j) - \frac{1}{n} \sum_{i=1}^{n} \varphi(Y_i) \right\}$$

Remarks:

- There is a way to simplify the objective for $\widetilde{\varphi}$ and find an approximate ERM. We will discuss it in the end
- The joint distribution of (Z_i, Y_i) may differ from π^*, but the objective includes only marginals
- A similar empirical functional (with $\mathcal{T}_T$ replaced by a different integral operator) was considered in [Korotin et al., 2024]

Theoretical analysis: assumptions

1. The density p_0 has a bounded support: $\|\Sigma^{-1/2}(x-m)\| \leqslant R$ for all $x \in \mathrm{supp}(\mathrm{p}_0)$

2. The density p_T has sub-Gaussian tails: $\mathbb{E}_{Y\sim \mathrm{p}_T} e^{u^\top Y} \leqslant e^{\mathrm{v}^2\|u\|^2/2}$ for any $u \in \mathbb{R}^d$

3. Every $\varphi \in \Phi$ satisfies $-L\|\Sigma^{-1/2}(x-m)\|^2 - M \leqslant \varphi(x) \leqslant M$ for all $x \in \mathbb{R}^d$

4. Normalization condition: for any $\varphi \in \Phi \cup \{\varphi^*\}$ it holds that $\mathcal{T}_\infty[\varphi] = 0$

5. The class Φ is parametric: $\Phi = \{\varphi_\theta : \theta \in \Theta\}$, where $\Theta \subseteq [-1,1]^D$

6. Lipschitz parametrization: there exists $\Lambda > 0$ such that

$$|\varphi_\theta(x) - \varphi_{\theta'}(x)| \leqslant \Lambda\left(1+\|x\|^2\right)\|\theta-\theta'\|_\infty \quad \text{for all } \theta,\theta' \in \Theta \text{ and all } x \in \mathbb{R}^d$$

Theoretical analysis: assumptions

- The assumption of compactly supported p_0 is usual (see, e.g., [Pooladian and Niles-Weed, 2024], [Korotin et al., 2024])
- In contrast to [Pooladian and Niles-Weed, 2024] and [Korotin et al., 2024], the target density p_T has an unbounded support
- The assumptions on φ_θ are satisfied in the cases of Gaussian marginals (see [Puchkin et al., 2025a, Proposition E.1]) and when log-potentials are neural networks with Lipschitz activations or logarithms of Gaussian mixtures

Theoretical analysis: motivation

[Korotin et al., 2024]: $\mathcal{O}(1/\sqrt{n})$ bound on the expected KL excess risk between couplings (based on Rademacher complexity)

[Pooladian and Niles-Weed, 2024]: squared TV-distance between path measures on $[0, \tau]$ scales as $\mathcal{O}\left(n^{-1/2} + (T-\tau)^{-k-2} n^{-1}\right)$, supports of p_0 and p_T are k-dimensional manifolds

Question. Is it possible to achieve $\mathcal{O}\left(\text{polylog}(n)/n\right)$ KL excess risk bounds?

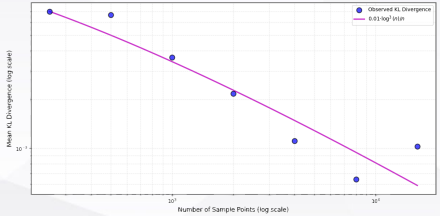


Figure: Mean KL excess risk in the case of Gaussian marginals (strategy B), $R^2 = 0.94$

Theorem: strategy A generalization error [Puchkin et al., 2025a]

Grant the aforementioned assumptions. Assume that b and T satisfy

$$bT \geqslant (5 + \log d) \vee \log\left(160 b \left(\mathrm{v}^2 \vee 1\right) \left\|\Sigma^{-1}\right\|\right)$$

Let $\widetilde{\mathrm{p}}_T$ be the density of $X_T^{\widetilde{\varphi}}$. Then, for any $\delta \in (0, 1/2)$, with probability at least $1 - 2\delta$, it holds that

$$\mathsf{KL}(\mathrm{p}_T, \widetilde{\mathrm{p}}_T) - \inf_{\varphi \in \Phi} \mathsf{KL}(\mathrm{p}_T, \mathrm{p}_T^{\varphi}) \lesssim \sqrt{\Omega(n, \delta) \inf_{\varphi \in \Phi} \mathsf{KL}(\mathrm{p}_T, \mathrm{p}_T^{\varphi})} + \Omega(n, \delta),$$

where

$$\Omega(n, \delta) = (Ld + M + d)\left(d + \log \frac{R\Lambda n}{\delta} + (M \vee \log L)\sqrt{d} e^{-bT}\right) \frac{D \log n}{n}$$

The hidden constant behind $\lesssim$ depends on Σ, m, b, and v only

Remark. The theorem admits $T = 1$. In this case, b should be sufficiently large

Theorem: strategy B generalization error [Puchkin et al., 2025b]

Grant the aforementioned assumptions and suppose that φ^* satisfies $-L\|\Sigma^{-1/2}(x-m)\|^2 - M \leqslant \varphi^*(x) \leqslant M$ for all $x \in \mathbb{R}^d$. For any $n \in \mathbb{N}$ and $\delta \in (0,1)$, introduce

$$\Upsilon(n,\delta) = D(M+d)\left(1 \vee \frac{L}{b}\right)\left(M + \log\left((L \vee \Lambda)nd\right) + \log(1/\delta)\right)\frac{\log n}{n} \lesssim \frac{D\log(n/\delta)\log n}{n}$$

and $\Delta = \inf_{\varphi \in \Phi} \mathrm{KL}(\pi^*, \pi^\varphi)$. Then there exists

$$T_0 \lesssim \frac{1}{b}\log\log\frac{1}{\Upsilon(n,1)} \vee \frac{1}{b}\log\left(d + LR^2 + \frac{Ld^2}{b} + M\right) \lesssim \log\log n$$

such that for any $\delta \in (0,1)$ and any $T \geqslant T_0$ with probability at least $(1-\delta)$ it holds that

$$\mathrm{KL}\left(\pi^*, \pi^{\widehat{\varphi}}\right) - \Delta \lesssim \sqrt{\Upsilon(n,\delta)\Delta\left(1 \vee \log\frac{(1 \vee L/b)(M+d)}{\Delta}\right)} + \Upsilon(n,\delta)\left(1 \vee \log\frac{(1 \vee L/b)(M+d)}{\Delta}\right).$$

Theoretical analysis: main results

Optimal control: $u^*(x,t) = \Sigma \nabla \log \mathcal{T}_{T-t}[e^{\varphi^*}](x)$

$$X_0 \sim \mathrm{p}_0, \quad \mathrm{d}X_t^* = -b(X_t^* - m)\mathrm{d}t + u^*(X_t^*, t)\mathrm{d}t + \Sigma^{1/2}\mathrm{d}W_t, \quad 0 < t < T$$

Estimate: $\widehat{u}(x,t) = \Sigma \nabla \log \mathcal{T}_{T-t}[e^{\widehat{\varphi}}](x)$

$$\widehat{X}_0 \sim \mathrm{p}_0, \quad \mathrm{d}\widehat{X}_t = -b(\widehat{X}_t - m)\mathrm{d}t + \widehat{u}(\widehat{X}_t, t)\mathrm{d}t + \Sigma^{1/2}\mathrm{d}W_t, \quad 0 < t < T,$$

Girsanov's theorem (see, e.g., [Domingo-Enrich et al., 2025, Theorem 2 and eq. (138)]):

$$\int_0^T \mathbb{E}_{X_t^*} \left\| \Sigma^{-1/2} \big(\widehat{u}(X_t^*, t) - u^*(X_t^*, t) \big) \right\|^2 \mathrm{d}t = 2\,\mathrm{KL}\left(\pi^*, \pi^{\widehat{\varphi}} \right)$$

Base process: $\mathrm{d}X_t^0 = b(X_t^0 - m)\mathrm{d}t + \sqrt{\varepsilon}\,\mathrm{d}W_t, \quad 0 < t < 1$

A simple way to construct baselines [Korotin et al., 2024]: take potentials of the form

$$e^{\varphi_\theta(y)} = e^{\|y\|^2/(2\varepsilon\sigma_1^2)} \sum_{k=1}^{K} \alpha_k \,\mathrm{p}(y; r_k, \varepsilon\sigma_1^2 S_k),$$

where $\sigma_1^2 = (1 - e^{-2b})/(2b)$ and $\mathrm{p}(\cdot; r_k, \varepsilon\sigma_1^2 S_k)$ is the density of $\mathcal{N}(r_k, \varepsilon\sigma_1^2 S_k)$

Remark 1. With such potentials $\log \mathcal{T}_t[e^{\varphi}]$ and $\pi^{\varphi}\left(y \,\middle|\, x\right)$ admit closed-form expressions

Remark 2. The choice of the Ornstein-Uhlenbeck process as a reference dynamics leads to better empirical results than [Korotin et al., 2024]

Lemma: [Puchkin et al., 2025a]

Let $\varphi : \mathbb{R}^d \to \mathbb{R}$ be such that $\varphi(x) \leqslant M$ for all $x \in \mathbb{R}^d$. Introduce

$$\mathcal{A}(x,t) = \left(\frac{be^2}{\sqrt{d}}\left\|\Sigma^{-1/2}(x-m)\right\|^2 + 4e^2\sqrt{d}\right)\arcsin(e^{-bt}) - 10e^2\sqrt{d}\log\left(1-e^{-2bt}\right)$$

and

$$\mathcal{K}(t) = \left(1-e^{-2bt}\right)^{-5e^2\sqrt{d}} \cdot \exp\left\{2e^2\sqrt{d}\arcsin(e^{-bt})\right\}$$

Then for any $x \in \mathbb{R}^d$ and $t \in [0, +\infty]$ it holds that

$$e^{-\mathcal{A}(x,t)\mathcal{K}(t)}\left(\frac{\mathcal{T}_\infty[e^\varphi]}{e^M}\right)^{\mathcal{K}(t)} \leqslant \frac{\mathcal{T}_t[e^\varphi](x)}{e^M} \leqslant e^{\mathcal{A}(x,t)/\mathcal{K}(t)}\left(\frac{\mathcal{T}_\infty[e^\varphi]}{e^M}\right)^{1/\mathcal{K}(t)}$$

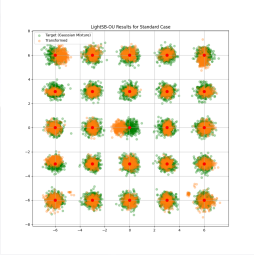
Implication:

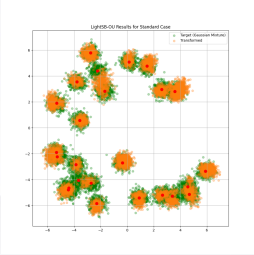
$$\begin{aligned}\log \mathrm{p}_T^{\varphi}(y) &= \log \int \frac{\mathrm{p}_0(x)\,\mathrm{q}_T(y\,|\,x)\,e^{\varphi(y)}}{\mathcal{T}_T[e^{\varphi}](x)}\,\mathrm{d}x \\ &\geqslant \log \int \mathrm{p}_0(x)\,\mathrm{q}_T(y\,|\,x)\,e^{\varphi(y)}\cdot\frac{e^{\mathcal{A}(x,T)/\mathcal{K}(T)}}{e^{M}}\left(\frac{e^{M}}{\mathcal{T}_\infty[e^{\varphi}]}\right)^{1/\mathcal{K}(T)}\mathrm{d}x \\ &\geqslant \varphi(y) - \frac{1}{\mathcal{K}(T)}\log \mathcal{T}_\infty[e^{\varphi}] + \underbrace{M\left(\frac{1}{\mathcal{K}(T)}-1\right)}_{\mathcal{O}\left(M\sqrt{d}\,e^{-bT}\right)} + \text{const.}\end{aligned}$$

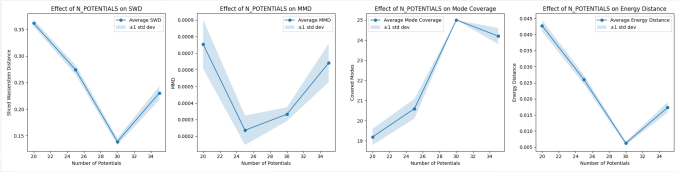
Simplified strategy A: take $bT = \Omega(\log n)$ and replace $\log \mathrm{p}_T^{\varphi}$ by its lower bound

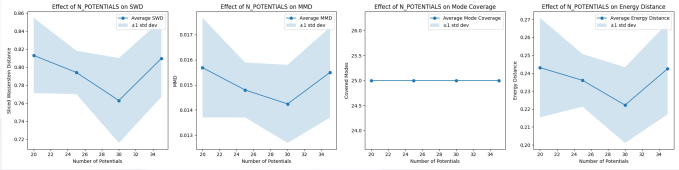
$$\check{\varphi} \in \operatorname*{Argmin}_{\varphi\in\Phi}\left\{\frac{1}{n}\sum_{i=1}^{n}\varphi(Y_i) - \frac{1}{\mathcal{K}(T)}\log \mathcal{T}_\infty[e^{\varphi}]\right\}$$

Evaluation on the Gaussian Mixture

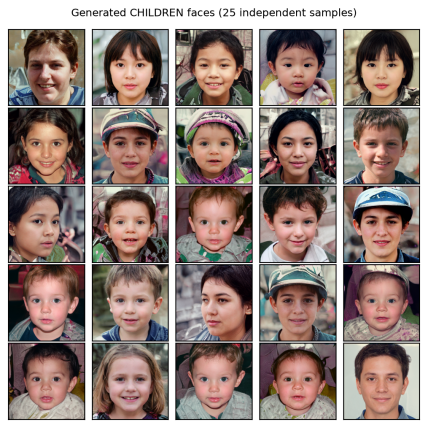








Examples I: 1024 ×1024 FFHQ dataset [Karras et al., 2019]



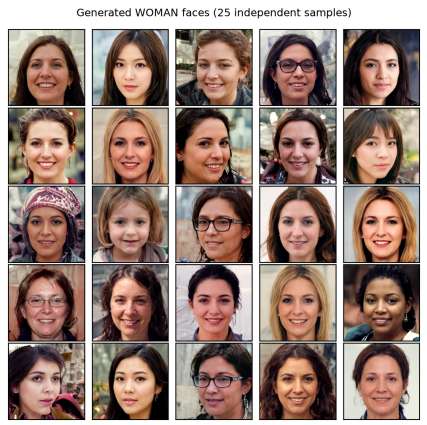


Figure: Examples of generated children (left) and women (right) faces based on 1024 × 1024 FFHQ dataset [Karras et al., 2019].

Further details are in our papers

D. Belomestny, A. Naumov, N. Puchkin, D. Suchkov
“Sample complexity of Schrodinger potential estimation”
arXiv:2506.03043 (to be updated soon)

N. Puchkin, D. Suchkov, A. Naumov, D. Belomestny
“Tight bounds for Schrödinger potential estimation in unpaired data translation”
ICLR 2026, arXiv:2508.07392

D. Belomestny, A. Naumov, N. Puchkin, D. Suchkov
“Schrödinger bridge problem via empirical risk minimization”
arXiv:2602.08374

References I

[Belomestny et al., 2026] Belomestny, D., Naumov, A., Puchkin, N., and Suchkov, D. (2026).
Schrödinger bridge problem via empirical risk minimization.
arXiv e-prints, page arXiv:2602.08374.

[Beurling, 1960] Beurling, A. (1960).
An automorphism of product measures.
Annals of Mathematics, 72(1):189–200.

[Dai Pra, 1991] Dai Pra, P. (1991).
A stochastic control approach to reciprocal diffusion processes.
Applied Mathematics and Optimization, 23(1):313–329.

References II

[Domingo-Enrich et al., 2025] Domingo-Enrich, C., Drozdzal, M., Karrer, B., and Chen, R. T. Q. (2025).
Adjoint matching: Fine-tuning flow and diffusion generative models with memoryless stochastic optimal control.
Preprint. ArXiv:2409.08861v5.

[Karras et al., 2019] Karras, T., Laine, S., and Aila, T. (2019).
A style-based generator architecture for generative adversarial networks.
In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4401–4410.

[Korotin et al., 2024] Korotin, A., Gushchin, N., and Burnaev, E. (2024).
Light Schrödinger bridge.
In *The Twelfth International Conference on Learning Representations*.

References III

[Leonard, 2014] Leonard, C. (2014).
A survey of the Schrödinger problem and some of its connections with optimal transport.
Discrete and Continuous Dynamical Systems, 34(4):1533–1574.

[Pooladian and Niles-Weed, 2024] Pooladian, A.-A. and Niles-Weed, J. (2024).
Plug-in estimation of Schrödinger bridges.
Preprint. ArXiv:2408.11686.

[Puchkin et al., 2025a] Puchkin, N., Pustovalov, I., Sapronov, Y., Suchkov, D., Naumov, A., and Belomestny, D. (2025a).
Sample complexity of schrödinger potential estimation.
Preprint. ArXiv:2506.03043.

[Puchkin et al., 2025b] Puchkin, N., Suchkov, D., Naumov, A., and Belomestny, D. (2025b).
Tight bounds for schrödinger potential estimation in unpaired data translation.
Preprint. ArXiv:2508.07392.

References IV

[Schrödinger, 1932] Schrödinger, E. (1932).
Über die Umkehrung der Naturgesetze.
Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse, pages 144–153.

Schrödinger equations:

$$\begin{cases} \mathrm{p}_0(x) = \nu_0(x) \int\limits_{\mathbb{R}^d} \mathrm{q}_T(y \,|\, x) \nu_T(y)\, \mathrm{d}y \\ \mathrm{p}_T(y) = \nu_T(y) \int\limits_{\mathbb{R}^d} \mathrm{q}_T(y \,|\, x) \nu_0(x)\, \mathrm{d}x \end{cases}$$

Introduce

$$g^\star(y) = \frac{\mathrm{p}_T(y)}{\nu_T(y)}$$

Then

$$\nu_0(x) \;=\; \frac{\mathrm{p}_0(x)}{\displaystyle\int q_T(x,z)\, \nu_T(z)\, dz} \;=\; \frac{\mathrm{p}_0(x)}{\displaystyle\int q_T(x,z)\, \frac{\mathrm{p}_T(z)}{g^\star(z)}\, dz}. \tag{6}$$

For each $x \in \mathbb{R}^d$ define

$$D_g(x) := \int_{\mathbb{R}^d} q_T(x, z)\, \frac{\mathrm{p}_T(z)}{g(z)}\, dz.$$

Substituting (6) into Schrödinger equations gives

$$\mathrm{p}_T(y) = \frac{\mathrm{p}_T(y)}{g^\star(y)} \int_{\mathbb{R}^d} \frac{q_T(x, y)\mathrm{p}_0(x)}{D_{g^\star}(x)}\, dx.$$

Canceling the common factor $\mathrm{p}_T(y)$ on both sides and multiplying by $g^\star(y)$, we obtain the nonlinear fixed–point equation

$$g^\star(y) = \int_{\mathbb{R}^d} \frac{q_T(x, y)\mathrm{p}_0(x)}{D_{g^\star}(x)}\, dx =: \mathcal{C}[g^\star](y)$$

The (static) Schrödinger bridge problem can be equivalently formulated as the problem of finding a positive function g solving the nonlinear fixed–point equation $g^\star = \mathcal{C}[g^\star]$.

Schrödinger bridge problem via empirical risk minimization

Assume that we are given independent samples

$$X_1, \dots, X_N \sim \mathrm{p}_0, \qquad Y_1, \dots, Y_M \sim \mathrm{p}_T,$$

Replacing ρ_0 and ρ_T by their empirical counterparts we obtain

$$\widehat{D}(x) := \frac{1}{M} \sum_{k=1}^{M} q_T(x, Y_k) \, \frac{1}{g(Y_k)} \quad \widehat{\mathcal{C}}_{N,M}[g](y) := \frac{1}{N} \sum_{i=1}^{N} \frac{q_T(X_i, y)}{\widehat{D}(X_i)}.$$

The fixed–point condition $g = \mathcal{C}[g]$ is approximated by enforcing

$$g(Y_j) \;\approx\; \widehat{\mathcal{C}}_{N,M}[g](Y_j), \qquad j = 1, \dots, M.$$

ERM:

$$\widehat{\mathcal{R}}_{N,M}(g) := \frac{1}{M} \sum_{j=1}^{M} \ell\Big(g(Y_j), \, \widehat{\mathcal{C}}_{N,M}[g](Y_j)\Big), \quad \widehat{g}_{N,M} \;\in\; \arg\min_{g \in \mathcal{G}} \widehat{\mathcal{R}}_{N,M}(g),$$

where $\mathcal{G}$ is an admissible class of positive functions or parametric models.

(Q) The reference kernel is Gaussian:

$$q_T(x,y) = (2\pi T)^{-d/2} \exp\left(-\frac{\|y-x\|^2}{2T}\right), \qquad x, y \in \mathbb{R}^d.$$

(R0) There exist $x_0 \in \mathbb{R}^d$ and two positive real numbers r_0, R_0 such that $B(x_0, r_0) \subseteq \operatorname{supp}(p_0) \subseteq B(x_0, R_0)$. Moreover, there is a constant $\rho_{0,-} > 0$ such that

$$p_0(x) \geqslant \rho_{0,-}, \qquad \forall x \in B(x_0, r_0).$$

(RT) There exist constants $0 < c_T^- \leqslant c_T^+ < \infty$ and $b_T^-, b_T^+ > 0$ such that

$$c_T^- \exp(-b_T^- \|y\|^2) \leqslant p_T(y) \leqslant c_T^+ \exp(-b_T^+ \|y\|^2), \qquad \forall y \in \mathbb{R}^d.$$

Recall that $g^\star = \mathcal{C}[g^\star]$ has the following form

$$g^\star(y) = \int_{\mathbb{R}^d} \frac{q_T(x,y)\mathrm{p}_0(x)}{D_{g^\star}(x)}\,dx = \int_{\mathbb{R}^d} q_T(y-x)w(x)\,dx\,,$$

Hence, $g^\star$ is a convolution. Fix $g^\star$, i.e. $\int \mathrm{p}_T(x)/g^\star(x)dx = \int \nu_T(x)dx = 1$.

Theorem: [Belomestny et al., 2026]

Assume conditions *(Q)*, *(R0)*, *(RT)* with $b_T^+ > 4/T$ and let $\ell(x,y) = (y-x)^2$. Then there exists a class $\mathcal{G}$ such that

$$\mathbb{E}[\mathcal{R}(\widehat{g}_{N,M})] \lesssim \log^{d/2}\max(M,N)\Big(N^{-1/2} + M^{-1/2}\Big)\,,$$

where $\lesssim$ stands for inequality up to constants independent of M and d and double logarithmic factors, and $\mathcal{R}(g) = \mathbb{E}_{\rho_T}[\ell(g,\mathcal{C}[g])]$.